

Section 1: Project Overview

Our project will become an open-access digital archive to show how space photography has evolved since the 1950s. Our collection directly contrasts grainy, historical glass film plates with high-definition, infrared imagery from modern telescopes like Hubble and James Webb. By sourcing public domain and Creative Commons assets from NASA, JPL, and NOIRLab, we are tracking the massive leap in visual clarity achieved once technology moved past Earth's atmospheric limitations.

Technically, this archive is built as a structured metadata project using CollectionBuilder-Sheets to convert historical imagery into clean, machine-readable data. Every asset is tracked using normalized fields like unique object IDs, exact capture dates, and specific telescope or observatory origins, ensuring the collection functions as an empirical database rather than a static photo album. Ultimately, this open-access structure makes the collection a versatile, real-world educational tool, allowing students, teachers, and independently curious astronomers to easily integrate the data into interactive timelines, study digital asset management, or analyze the technical evolution of how we map the universe.

Section 2: Roles & Responsibilities

Ben and I have both collaborated on all technical and editorial aspects of the project so far. While we both sourced historical assets and input technical data for our initial Metadata Draft, Ben focused on populating the bulk of the spreadsheet items. Aside from linking our metadata spreadsheet to our main repository, we have customized our repository with a personalized banner and unique layout.

For the written portions of the project, I took the lead on drafting the foundational text ahead of time before having Ben review and refine it. These written portions include our core project proposal, the formal project overview, the collection's theme descriptions, and the technical statement explaining how we transformed raw data into an educational archive. This workflow allowed us to balance the technical backend work on GitHub with peer-reviewed writing.

Section 3: Data Inventory & Provenance

Source objects

Source Institution	# Objects	Accessed by	Rights
NASA	15	https://www.nasa.gov/	https://www.nasa.gov/nasa-brand-center/images-and-media/
NSF NOIR Lab	2	https://noirlab.edu/public/	https://noirlab.edu/public/pressroom/
ESA Hubble Gallery	1	https://esahubble.org/	ESA/Hubble Copyright and Image Usage Policy
NASA ISS and JSC	1	https://eol.jsc.nasa.gov/	https://www.nasa.gov/nasa-brand-center/advertising-guidelines/
Smithsonian Museum	1	https://www.smithsonianmag.com/air-space-magazine/the-first-photo-from-space-13721411/	https://library.si.edu/copyright

File inventory

We are including mostly JPGs, PNGs, and Webp files to easily access the photos on the website.

Tools and access

Our project is an analysis of data management frameworks to build a professional-grade digital repository to show the history of Space Imaging. The technical foundation of our archive relies on a cloud-based workflow where a highly structured Google Sheet serves as our primary metadata inventory. Cataloging every single item with standardized fields for critical information while integrating a master data spreadsheet into a GitHub repository using the CollectionBuilder framework will create a searchable website with an extensive gallery that is also easy to grow and maintain. Because this is a public digital humanities project, we are also focusing on digital

ethics and intellectual property. We are strictly using public domain assets from government entities like JPL and NASA, and being mindful of other possible factors like Creative Commons licenses for the European Space Agency (ESA) and ESO.

Section 4: Storage & Stewardship

During the project

All of our files are stored in our Google Drive folder as well as our main GitHub repository. The two of us, Ben Johnson and Jackson Woolfolk, are the only two that have access to the Google Drive folder. The Images and metadata have been cataloged on a Google Sheets page that is directly linked to the main repository, and is called on in the website to provide the Item's image and data. Our metadata spreadsheet is public and available for anyone to access.

After term

We would hope that this project will last forever, but of course, without the proper monitoring and maintenance, it would likely crash or rot on the internet and become unusable in the distant future. Maintaining this website for the future would likely include a lot of updating, both with new information and new items being released.

Section 5: Access, Reuse & Rights

Your license

The license we have chosen for our created content is the Creative Commons license. This will allow others to share, use or build upon our work.

Rights implications of your sources

All of the items used in our collection are in the public domain; those rights will also dictate what others are allowed to do with items in our collection.

Ethical considerations

In our collection we have two images of individuals Buzz Aldrin and Bruce McCandless that have the potential for ethical considerations. Most of the images in our collection are of celestial bodies with photos taken from spacecraft which avoids ethical issues.

Appendix: Data Dictionary

For our file-naming convention, we aimed to keep it simple. We included keywords for what was in the photo being used, what instrument was used to take the photo, and any other standout details. We also included an 'S' or 'G' in the file name to indicate whether the photo was taken in space or from Earth. Most of our files had unique convenient titles; we didn't need to come up with our own unique ones. As we were choosing photographs of celestial bodies, it was clear what we would need to include in our subjects section.

Field Name	Definition	Example Value	Notes on Use
Object ID	A unique identifier is given to each individual item in the collection so they can be easily tracked.	coll02	Although all of our items already have unique filenames, giving them a simple ID makes sorting even simpler.
Filename	The exact name of the digital file stored on the computer or server.	sun_s_skylab.jpg	Each file includes an 'S' or 'G' to designate if the photo came from 'Space' or 'Ground.'
Title	The official name of the item used for display on the website, giving viewers a quick idea of what they are looking at.	Occultation of Mars	Having an official title makes the site more appealing than identifying items by filename.
Format	The physical or digital nature of the item, whether a picture, a video, or a document.	image/jpeg	To track what formats are being included, since we are restricted to PNG and JPG.
Date	The specific day, month, or year the original item was created or captured.	06-04-1998	We consolidated the dates into the same format and avoided complicated formats.

Subject	A short description of what the item has captured.	Crescent Nebula	We found that having an official subject line made it easier for readers to view.
Keywords	Keywords or tags that describe the main topics, themes, or objects featured in the item to make searching easier.	'Mercury', 'Sun', 'Eclipse', 'Space'	Originally just 'Subject,' adding keywords will allow photos to be grouped according to what someone is looking for.
Location	An approximation or exact location of where an observatory or telescope captured the photo.	Gusev Crater, Mars	We tried to detail photos taken from space with an approximate location and mission when we could.
Image Link	The URL or web path that directs users to the source where the digital file was originally found.	https://science.nasa.gov/image-detail/hubble-space-telescope-hst/	We took links directly from where we first found the photo, which brought us through multiple different galleries.
Rights / Rights Statement	A formal legal statement indicating the copyright status of the item and explaining how others are allowed to use or share it.	https://www.nasa.gov/nasa-brand-center/images-and-media/	We included both the NASA official use-rights webpage and the US Copyright website to show our compliance.
Creator	The name of the organization, photographer, artist, or agency responsible for making the original item.	NASA / Apollo 13 Crew	A way to include the name of a creator, even though most items were captured with an automated telescope.
Description	A short paragraph or summary that provides context, details the history, and explains what is	Ground-based image taken of the Crescent Nebula	To give some context to the massive and unknown figures being captured on camera.

	happening in the image or document.		
Taken by	The specific telescope equipment or operator who physically captured the image or recorded the data	Skylab 4	Similar to the creator, meant to detail the satellite, telescope, or mission that took a photo.